

HouseHunt

House Rental Application

Solution Architecture Document

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Project Name	HouseHunt – House Rental Application
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1. Introduction

HouseHunt is a powerful and user-centric house rental application designed to revolutionize the way people engage with the real estate market. Whether searching for a home to rent, planning to sell a property, or working as a real estate professional, HouseHunt offers a seamless, all-in-one platform to meet diverse needs.

With a focus on simplifying the often overwhelming and time-consuming process of property transactions, the app provides users with detailed listings, real-time market insights, and comprehensive property data to support well-informed decision-making. Its advanced search capabilities, virtual tours, and end-to-end transaction facilitation make HouseHunt a reliable and convenient digital marketplace for all real estate stakeholders.

2. Project Overview

Project Summary	
Application Name	HouseHunt
Type	House Rental & Real Estate Marketplace
Platform	Web & Mobile (Cross-platform)
Primary Users	Renters, Landlords, Buyers, Sellers, Real Estate Agents
Availability	24/7 across all devices

3. Solution Objectives

The HouseHunt solution architecture is designed to achieve the following core objectives:

- Provide a centralized, scalable platform that connects renters, buyers, landlords, sellers, and real estate agents.
- Enable powerful property discovery through advanced search, filtering by location, price, and amenities.
- Deliver real-time market insights and property data to support well-informed decision-making.
- Facilitate the entire real estate journey — from listing and search to negotiation and paperwork.
- Offer virtual tour capabilities to enhance remote property evaluation.
- Ensure 24/7 accessibility across web and mobile devices with a responsive, intuitive interface.
- Promote transparency through accurate, verified property information and user reviews.
- Serve as a marketing and client outreach tool for real estate agents and brokers.

4. Proposed Solution

HouseHunt adopts a modern, cloud-native microservices architecture to deliver a scalable, maintainable, and high-performance platform. The solution is divided into clearly defined functional layers, each addressing a specific aspect of the real estate marketplace experience.

4.1 Architecture Overview

The proposed architecture follows a three-tier model — Presentation Layer, Application/Business Logic Layer, and Data Layer — deployed on a cloud infrastructure (AWS/Azure) to ensure high availability, elasticity, and security.

Presentation Layer	Application Layer	Data Layer
React.js Responsive Web UI Virtual Tour Viewer.	API Gateway Auth Service Search Service Booking.	MongoDB (Document) AWS S3 (Media Storage)

4.2 Core Modules & Microservices

4.2.1 User Management & Authentication Service

- Role-based access control (RENTER, Owner, ADMIN)
- OAuth 2.0 / JWT-based secure authentication

4.2.2 Property Listing Service

- Comprehensive listing creation with multi-image upload and video support

- Property categorization: rental, sale, commercial, residential
- Listing verification and approval workflow for administrators

4.2.4 Communication & Booking Service

- In-app messaging between renters and landlords / agents
- Property visit scheduling and calendar integration
- Rental agreement generation and e-signature support
- Payment gateway integration for security deposits and rent payments

4.2.5 Notification Service

- Push notifications for new listings matching saved searches
- Email and SMS alerts for booking confirmations, messages, and reminders
- In-app notification centre with read/unread tracking

5. Solution Architecture Diagram

The diagram below illustrates the high-level architecture and data flow of the HouseHunt application, depicting interaction between users, front-end clients, backend microservices, third-party integrations, and the data layer.

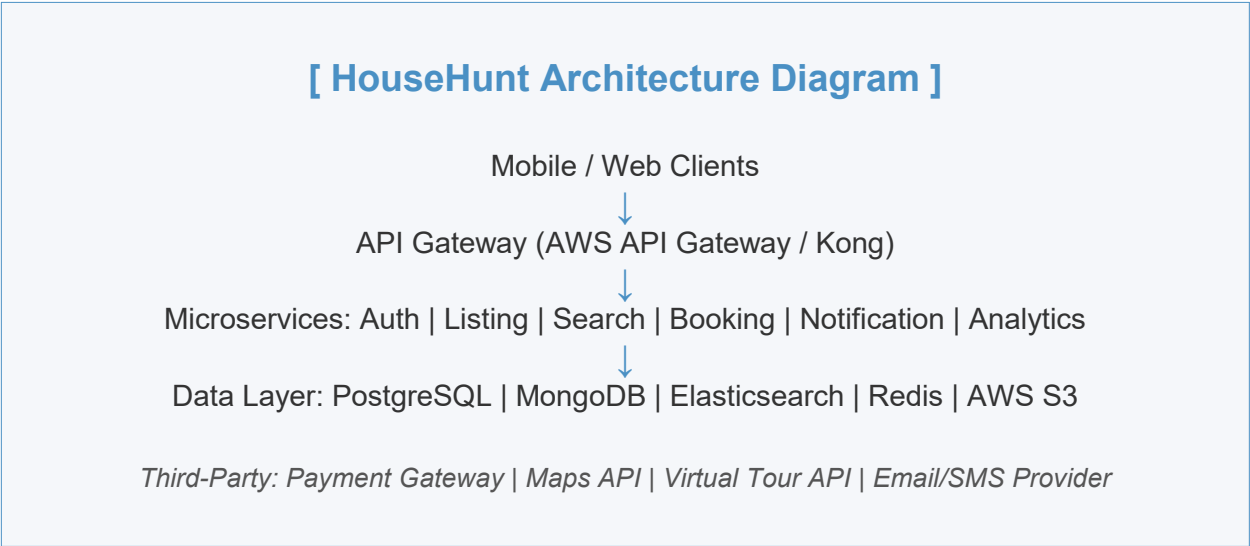


Figure 1: High-Level Solution Architecture and Data Flow – HouseHunt Application

6. Technology Stack

Recommended Technology Stack

Frontend (Web)	React.js, Tailwind CSS
Backend	Node.js (Express) / Python (FastAPI)
API Layer	RESTful APIs
Document Store	MongoDB (property listings, user profiles)
Cache Layer	Redis (sessions, search results)
Media Storage	Cloudinary
Authentication	OAuth 2.0, JWT, Firebase Auth
CI/CD	GitHub Actions, Docker, Kubernetes

7. Risks & Mitigations

Risk	Impact	Mitigation Strategy
Data breach / unauthorized access	High	End-to-end encryption, RBAC, regular penetration testing
Third-party API downtime (Maps, Payment)	Medium	Fallback providers, circuit breaker pattern, retries
Scalability bottlenecks during peak load	High	Auto-scaling policies, Redis caching, CDN for static assets
Fraudulent property listings	High	Admin verification workflow, AI-based fraud detection, user reporting
Regulatory non-compliance (GDPR/IT Act)	High	Built-in consent management, data residency controls, legal review
High initial development cost	Medium	Phased delivery roadmap; MVP-first approach with iterative releases

8. Conclusion

The proposed solution architecture for HouseHunt establishes a robust, scalable, and secure foundation for a modern real estate marketplace. By leveraging a cloud-native microservices approach, the platform is well-positioned to handle high user concurrency, complex search scenarios, and diverse real estate transaction workflows.

The phased delivery model — beginning with an MVP covering core listing, search, and communication features — ensures early value delivery while laying the groundwork for advanced capabilities such as AI-powered recommendations, market analytics, and virtual property tours. With transparent information, 24/7 availability, and an intuitive experience across devices, HouseHunt is poised to become the go-to digital platform for all real estate stakeholders.

End of Solution Architecture Document | HouseHunt v1.0